

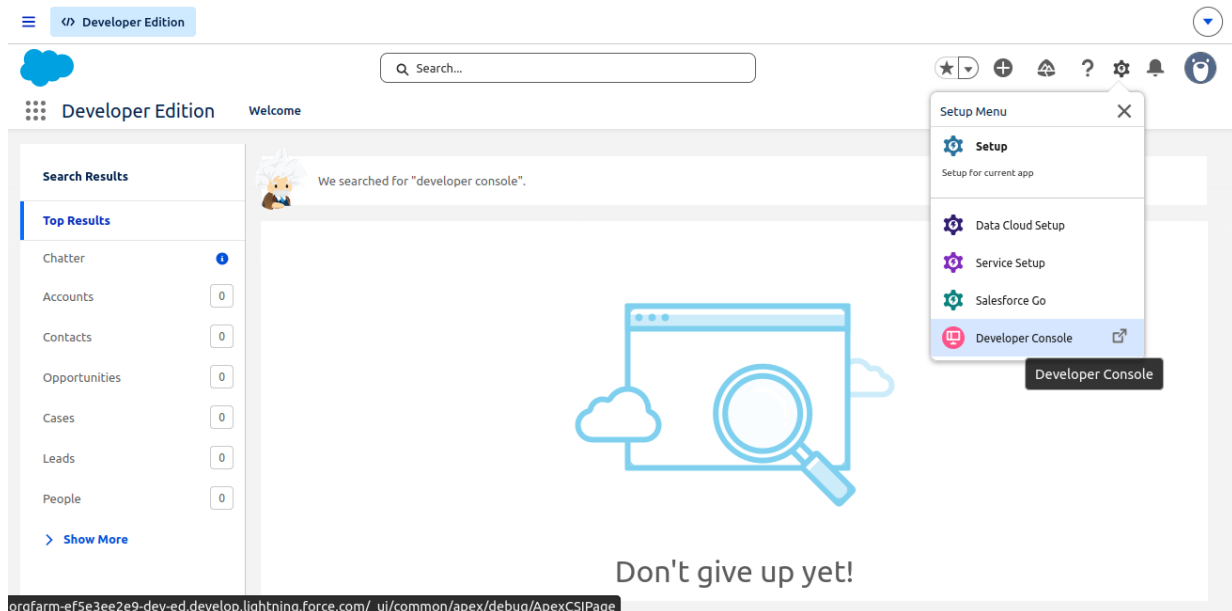
Step-by-Step Procedure: Salesforce Calculator Project

Step 1: Login to Salesforce

1. Open your browser
2. Go to Salesforce login page
3. Enter your **username and password**
4. Click **Login**

Step 2: Open Developer Console

1. After login, go to the **top right corner**
2. Click on the **Settings (⚙️ Gear icon)**
3. Click on **Developer Console**
4. A new window (Developer Console) will open



Step 3: Create Apex Controller Class

1. Click on **File**
2. Click **New → Apex Class**
3. Enter class name: **CalculatorVFController**
4. Click **OK**
5. Paste controller code and click **Save**

```
public class CalculatorVFController {
```

```
public Decimal num1 {get; set;}
```

```
public Decimal num2 {get; set;}
```

```
public Decimal result {get; set;}
```

```
// Basic Operations
```

```
public void add() {
```

```
    result = num1 + num2;
```

```
}
```

```
public void subtract() {
```

```
    result = num1 - num2;
```

```
}
```

```
public void multiply() {
```

```
    result = num1 * num2;
```

```
}
```

```
public void divide() {
```

```
    if(num2 == 0) {
```

```
        ApexPages.addMessage(new ApexPages.Message(ApexPages.Severity.ERROR, 'Cannot  
divide by zero'));
    }
```

```
    } else {
```

```
        result = num1 / num2;
```

```
    }
```

```
}
```

```
// Scientific Operations
```

```
public void square() {  
    result = num1 * num1;  
}
```

```
public void squareRoot() {  
    result = Decimal.valueOf(Math.sqrt((Double)num1));  
}
```

```
public void power() {  
    result = Decimal.valueOf(Math.pow((Double)num1, (Double)num2));  
}
```

```
public void sinValue() {  
    result = Decimal.valueOf(Math.sin((Double)num1));  
}
```

```
public void cosValue() {  
    result = Decimal.valueOf(Math.cos((Double)num1));  
}
```

```
public void tanValue() {  
    result = Decimal.valueOf(Math.tan((Double)num1));  
}
```

```
public void logValue() {  
    result = Decimal.valueOf(Math.log10((Double)num1));  
}
```

```
}  
  
}
```

Step 4: Create Visualforce Page (UI)

1. Click on **File**
2. Click **New** → **Visualforce Page**
3. Enter page name: CalculatorPage
4. Click **OK**
5. Paste Visualforce code and click **Save**

```
<apex:page controller="CalculatorVFController">
```

```
<apex:form >
```

```
<apex:pageBlock title="Scientific Calculator">
```

```
<apex:pageMessages />
```

```
<!-- Input Section -->
```

```
<apex:pageBlockSection title="Enter Values" columns="1">
```

```
<apex:inputText value="{!num1}" label="Number 1" required="true"/>
```

```
<apex:inputText value="{!num2}" label="Number 2 (for Power)" />
```

```
</apex:pageBlockSection>
```

```
<!-- Basic Operations -->
```

```
<apex:pageBlockSection title="Basic Operations" columns="4">
```

```
<apex:commandButton value="+" action="{!add}" rerender="res"/>
```

```
<apex:commandButton value="-" action="{!subtract}" rerender="res"/>
```

```
<apex:commandButton value="×" action="{!multiply}" rerender="res"/>
```

```
<apex:commandButton value="÷" action="{!divide}" rerender="res"/>
```

`</apex:pageBlockSection>`

`<!-- Scientific Operations -->`

`<apex:pageBlockSection title="Scientific Functions" columns="4">`

`<apex:commandButton value="x2" action="{!square}" rerender="res"/>`

`<apex:commandButton value="√x" action="{!squareRoot}" rerender="res"/>`

`<apex:commandButton value="xy" action="{!power}" rerender="res"/>`

`<apex:commandButton value="log" action="{!logValue}" rerender="res"/>`

`<apex:commandButton value="sin" action="{!sinValue}" rerender="res"/>`

`<apex:commandButton value="cos" action="{!cosValue}" rerender="res"/>`

`<apex:commandButton value="tan" action="{!tanValue}" rerender="res"/>`

`</apex:pageBlockSection>`

`<!-- Result -->`

`<apex:outputPanel id="res">`

`<apex:pageBlockSection title="Result">`

`<apex:outputText value="{!result}" style="font-size:20px; font-weight:bold; color:blue;"/>`

`</apex:pageBlockSection>`

`</apex:outputPanel>`

`</apex:pageBlock>`

`</apex:form>`

`</apex:page>`

Step 5: Preview the Application

1. In Developer Console, open CalculatorPage

2. Click on **Preview** (top right corner)
3. A new browser tab will open

The screenshot shows a web-based scientific calculator interface. At the top, there's a title bar labeled "Scientific Calculator". Below it, a section titled "Enter Values" contains two input fields: "Number 1" with the value "10" and "Number 2 (for Power)" with the value "2". The next section, "Basic Operations", displays four buttons: "+", "-", "x", and "/". Below that, the "Scientific Functions" section includes buttons for "x²", "√x", "xʸ", "log", "sin", "cos", and "tan". The final section, "Result", shows the calculated value "100" in a large, bold, blue font.

Step 6: Test the Calculator

1. Enter value in **First Number** field
2. Enter value in **Second Number** field
3. Click any operation:
 - Add
 - Subtract
 - Multiply
 - Divide
4. View the result displayed on screen.

Conclusion

The calculator application is successfully created using Apex and Visualforce. The system performs arithmetic operations and displays results dynamically.